

Predicting Enforced Disappearances: A Machine Learning Analysis of Temporal and Network Features

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Abstract

Enforced disappearances are a severe human rights violation, but their mechanisms remain contested. Do they reflect actor-network dynamics, such as fragmentation and coalitions, or do they result from state institutional capacity operating locally? We use 7 years of georeferenced conflict event data—202,347 events from Nigeria, Mexico, and Myanmar (2018–2024)—and machine learning (ML) to evaluate these theoretical accounts. Despite the extreme rarity of disappearances (2.9% of events), LSTM models perform strongly, achieving an Area Under the Precision-Recall Curve (AUPRC) of 0.745. Systematic ablation studies test network versus institutional indicators. We find that twelve graph-theoretic features—density, centralization, clustering—provide zero predictive value (0.6% contribution). In contrast, simple event-based features such as arrests, violence, and fatalities drive performance. This negative result carries substantive implications. Disappearances are local temporal phenomena driven by state repression capacity and conflict intensity, rather than by actor network structure or spatial contagion. The models also predict new outbreaks even in the absence of prior disappearance (AUPRC = 0.666). Past violations contribute 10.1% to performance, which indicates systematic patterns beyond stochastic violence. These findings challenge network-centric theories of repression and establish the computational feasibility of early warning systems targeting severe but rare violations.

Keywords: Enforced disappearances, Conflict prediction, Machine learning, Network analysis, State repression, Early warning systems

1 Introduction

Two broad theoretical traditions offer foundational explanations for patterns of violence in armed conflict. The first, rooted in network-centric theories of conflict dynamics [1, 2], emphasizes how actor relationships shape violence through alliance formation, fragmentation, and the relational structure among armed groups. Although this literature does not specifically address enforced disappearances, it implies that patterns of violence may depend on who is connected to whom, suggesting that network structure, centrality, and clustering could help predict conflict outcomes. We therefore examine whether these relational mechanisms extend to disappearances. The second tradition draws on theories of state capacity and repression [3–5], which conceptualize severe forms of violence as products of institutional power and bureaucratic organization. While not disappearance-specific, this perspective implies that such violations are more likely where states possess the territorial control and administrative capacity necessary to detain, conceal, and deny responsibility. From this view, disappearances should follow observable manifestations of state presence and conflict intensity rather than actor network configurations.

These theoretical accounts offer diverging implications for understanding the micro-mechanisms of state terror. If network structure drives disappearances, interventions should target actor fragmentation and coalition dynamics. Conversely, if institutional capacity is the primary driver, efforts should focus on accountability mechanisms and territorial governance. Existing empirical research has largely incorporated both network and institutional variables without explicitly evaluating their distinctive predictive contributions.

This paper leverages Machine Learning (ML) as a theory-testing tool to adjudicate between these accounts. By systematically comparing models with and without network structure features, we can assess whether actor relational patterns provide a predictive signal beyond institutional and event-based indicators. Our approach follows recent methodological developments using prediction tasks to evaluate social theories [6, 7]: if network theories are correct, network features should improve forecast accuracy; if institutional capacity dominates, simple event-based features should suffice. By mapping competing theoretical expectations onto distinct feature sets and evaluating their predictive contributions, we treat model comparison as a direct empirical test of these explanations.

Enforced disappearances (the arrest, detention, or abduction of persons by state agents or organized groups followed by a refusal to acknowledge their fate) represent one of the most severe human rights violations in armed conflict [8]. Unlike other forms of violence that leave immediate evidence, disappearances are characterized by intentional concealment, creating profound uncertainty for families and communities. The practice serves as a tool of intimidation and social control, generating fear that extends far beyond direct victims.

Despite their severity, disappearances have received limited attention in computational conflict prediction research. Existing early warning systems typically focus on broader categories such as battle deaths, civilian targeting, or general violence. This aggregation obscures important distinctions: the mechanisms driving disappearances may differ fundamentally from those producing other conflict outcomes. Understanding and predicting disappearances as a distinct phenomenon is essential for targeted monitoring, advocacy, and protection efforts.

Predicting enforced disappearances presents substantial methodological challenges. In our dataset of 202,347 conflict events spanning 2018–2024, only 5,959 (2.9%) involve disappearances. This extreme class imbalance creates difficulties for standard ML approaches, which may achieve high accuracy by simply predicting the majority class while failing to identify positive cases. The rarity also reflects the nature of the phenomenon: disappearances are not randomly distributed but occur in specific contexts characterized by state repression, weak accountability, and particular conflict dynamics. Effective prediction requires distinguishing true precursors from the vast background of non-disappearance events, identifying subtle patterns that precede these relatively uncommon occurrences.

1.1 Research Questions and Approach

This study addresses three primary questions:

1. Can enforced disappearances be predicted at operationally useful accuracy levels despite their rarity (2.9% of events)?
2. Do actor network features improve prediction beyond simple event-based indicators?
3. Can models predict new outbreaks in regions without a disappearance history, or do they rely primarily on autocorrelation (recent disappearances predicting future disappearances)?

To address these questions, we employ temporal LSTM models with six-month lookback windows applied to conflict event data from Nigeria, Mexico, and Myanmar (2018–2024). Our approach utilizes systematic ablation studies across nine feature configurations to evaluate the relative importance of different data types. Specifically, we compare the predictive signal provided by network structure metrics (such as density, centralization, and clustering) against fundamental event-based features, including violence counts, arrests, and fatalities. This framework allows us to adjudicate between theoretical drivers of disappearance patterns while testing the models’ capacity to identify new outbreaks in historically unaffected regions.

1.2 Contributions

This research makes three primary contributions:

First, our analysis indicates that enforced disappearances align more closely with institutionally driven phenomena than with actor-network dynamics. Through comparative modeling, we show that graph-theoretic features derived from actor co-occurrence provide minimal additional predictive utility beyond fundamental

event-based indicators. This finding is theoretically informative: it suggests that disappearances primarily reflect local state capacity and conflict intensity rather than the relational configurations of armed actors. The limited impact of network features provides empirical support for state capacity explanations of repression, suggesting these violations are driven by bureaucratic apparatus rather than network diffusion.

Second, we demonstrate that disappearances exhibit strong, but not exclusive, temporal autocorrelation, confirming their genuine early-warning potential. While the persistence of repressive capacity makes past events a strong predictor, our models successfully identify emerging disappearance risk in regions without a prior history of such violations. By distinguishing between established patterns and new outbreaks, we show that predictive systems can detect the onset of repression before it becomes entrenched, shifting the utility of monitoring from tracking ongoing abuses to enabling preventive interventions.

Third, we establish that rare, severe human rights violations are computationally tractable targets despite extreme class imbalance. By applying rigorous evaluation metrics and temporal validation to a dataset in which disappearances represent a tiny fraction of total events, we demonstrate that low-prevalence outcomes can be identified as distinct phenomena. This methodological contribution demonstrates that specific types of violations—often obscured when aggregated into general conflict categories—can be monitored with precision, allowing for more targeted and specialized human rights forecasting.

The remainder of this paper proceeds as follows: Section 2 reviews related work, Section 3 describes data and methods, Section 4 presents results, and Section 5 discusses implications and limitations.

2 Related Work

Enforced disappearances represent a critical yet computationally understudied human rights violation. This section positions our work within four key research streams: conflict-prediction methodologies, the rare-event challenge, temporal modeling approaches, and network-based conflict analysis. We demonstrate that while existing research provides methodological foundations, no prior work specifically targets disappearance prediction, leaving a significant gap that we address.

2.1 Conflict Prediction and Early Warning Systems

Modern conflict forecasting has evolved from structural models emphasizing slow-moving variables toward dynamic event-based approaches leveraging ML. The ViEWS (Violence Early Warning System) consortium represents the state-of-the-art in sub-national violence prediction, forecasting state-based conflict and one-sided violence at country-month and PRIO-GRID-month levels using ensemble methods combining random forests, hurdle models, and deep learning architectures [9, 10]. Their 2023/24 prediction challenge revealed a critical limitation: even sophisticated models struggle to forecast conflict onset in previously peaceful locations, often performing worse than simple “no-change” baselines [11]. This finding motivates our focus on distinguishing genuine early warning from temporal autocorrelation.

Event-based forecasting systems, such as ICEWS (Integrated Crisis Early Warning System), process millions of machine-coded events from news sources to generate short-term forecasts [12]. However, these systems predominantly target aggregate outcomes — battle deaths, protest escalation, regime instability — rather than specific human rights violations. The Political Instability Task Force achieves over 80% accuracy in predicting state fragility using minimal structural variables [13], but operates at the annual country-level granularity, unsuitable for operational protection efforts. Our subnational monthly prediction of disappearances fills this temporal and substantive gap.

Methodologically, conflict prediction faces a fundamental tension between statistical significance and predictive accuracy. Ward et al. [14] demonstrate that influential civil war models correctly predict only one of 107 actual onsets despite statistically significant coefficients, cautioning against “policy by p-value.” Blair and Sambanis [15] argue that process variables (recent events, short-term changes) can substitute for structural variables over brief forecasting horizons, supporting our event-based approach. However, Chadeaux and Schincariol [16] show that autoregressive models consistently outperform complex alternatives, with structural covariates frequently degrading performance—a finding that informs our ablation study design testing whether network complexity adds value.

Recent work applying deep learning to conflict contexts remains limited. Radford [17] employs Convolutional LSTM combining spatial convolutions with temporal memory, while Malone [18] uses standard RNNs for escalation forecasting. Both achieve competitive performance but rely heavily on lagged fatality levels, struggling to forecast new escalations. Our contribution extends this temporal modeling tradition to a previously unexamined outcome: enforced disappearances.

2.2 Enforced Disappearances: A Computational Gap

Enforced disappearances—defined by the International Convention as deprivation of liberty by state agents or with their acquiescence, followed by concealment of fate [8]—constitute a distinct violation requiring separate analytical treatment. Unlike killings that leave bodies, disappearances create prolonged uncertainty serving dual purposes: eliminating opponents and terrorizing communities through ambiguity [19]. The UN Working Group on Enforced or Involuntary Disappearances documents over 56,000 cases across 112 states since 1980, with 45,000 fates still unknown [20].

Our study countries exemplify different contexts of disappearance. Mexico faces 115,000+ cases since 1962, with 90% occurring after 2006 as organized crime and state forces colluded during militarized anti-drug operations [21]. Myanmar experienced systematic disappearances following the 2021 coup, targeting activists, journalists, and opposition politicians as part of broader repression [22]. Nigeria’s disappearances occur across multiple conflict zones involving Boko Haram, ISWAP, and security force operations in the Northeast and Southeast [23].

Existing disappearance research is overwhelmingly qualitative, focusing on legal accountability [24], transitional justice mechanisms, and victim advocacy. The Human

Rights Data Analysis Group (HRDAG) conducts sophisticated statistical work on disappearances but emphasizes population estimation via Multiple Systems Estimation rather than prediction [25].

Critical gap: No published computational work applies ML specifically to predict enforced disappearances. Mueller and Rauh [26] use newspaper text and topic modeling to forecast political violence, and recent work predicts state violence in Mexico and Colombia [27], but these studies aggregate violence types rather than isolating disappearances. Our work is the first to treat disappearances as a distinct prediction target, addressing this significant lacuna.

2.3 Rare Events in Conflict: Methodological Foundations

Predicting disappearances poses the rare-event challenge: only 2.9% of conflict events in our data involve disappearances, resulting in severe class imbalance. King and Zeng [28, 29] establish that standard logistic regression produces biased estimates for rare outcomes, systematically underestimating event probability. Their bias-correction methods have become foundational for conflict research addressing low-prevalence outcomes, such as civil war onset (1-2% prevalence) [14].

ML approaches offer advantages for imbalanced classification. Muchlinski et al. [30] demonstrate that random forests substantially outperform logistic regression for civil war onset (1.8% prevalence), achieving an AUROC of 0.91, whereas standard logit fails to identify any onsets. He and Garcia [31] provide a comprehensive review of imbalanced learning challenges: classifiers’ bias toward the majority class, decision boundaries favor frequent patterns, and small minority-class clusters (“small disjuncts”) are ignored.

Evaluation metric selection proves critical. Saito and Rehmsmeier [32] demonstrate that ROC curves can be misleadingly optimistic for imbalanced data because the False Positive Rate denominator includes vast numbers of true negatives. Precision-Recall curves better reflect operational performance when minority class identification is the primary goal. For 2.9% prevalence, random classifier AUPRC baseline is 0.029, while AUROC baseline remains 0.50 regardless of imbalance—making AUPRC more informative. We follow this guidance by reporting AUPRC as our primary metric.

Temporal autocorrelation creates an additional methodological challenge in conflict prediction. Hegre et al. [33] show that countries exiting civil war face a 44% recurrence probability within five years, with conflict-trap effects persisting for 25+ years [34]. Chadeaux and Schincariol [16] argue that autoregressive models are “not merely convenient baselines but theoretically appropriate tools” because conflict is driven by internal feedback and short-term adaptation rather than slow-changing structural conditions. They show that adding structural covariates to autoregressive models yields only 0.2% improvement, while covariates alone perform 30% worse.

This creates a validation challenge: distinguishing genuine forecasting from persistence-based pattern matching. Our ablation study, which removes past disappearances, directly addresses this by testing whether models retain predictive power for new outbreak regions—what Mueller and Rauh [35] term the “hard problem” of conflict prevention.

2.4 Temporal Modeling for Conflict Forecasting

Long Short-Term Memory (LSTM) networks have emerged as promising architectures for conflict prediction due to their ability to capture long-range temporal dependencies. Beck, King, and Zeng [36] first documented neural network advantages for international conflict, arguing that conflict causes are “highly non-linear, massively interactive, and heavily context dependent”—properties well-suited to deep learning.

Recent applications demonstrate mixed results. Malone [18] applies standard RNNs to conflict escalation, finding they effectively learn “conflict traps” (violence persistence) but struggle with onset prediction in peaceful locations. Radford [17] achieves strong performance with Convolutional LSTM combining spatial and temporal features, but acknowledges heavy reliance on lagged fatality levels. Von der Maase [37] develops a state-of-the-art HydraNet architecture (Monte Carlo Dropout LSTM U-Net), emphasizing that “while conflict is inert, it is not static”—models must capture nuanced temporal structure beyond simple autocorrelation.

Graph neural networks extend temporal modeling to capture spatial dependencies. Brandt et al. [38] employ Spatio-Temporal Graph Convolutional Networks (ST-GCN), improving over ViEWS benchmarks by modeling spatial autocorrelation alongside temporal patterns. However, performance gains remain modest, and models still rely predominantly on conflict history.

Attention mechanisms offer interpretability advantages. Temporal attention weights reveal which months in a sequence contribute most to predictions, providing transparency valuable for policy applications [39]. We implement attention-LSTM to enable analysis of temporal importance patterns in disappearance forecasting.

A critical finding from recent prediction competitions is that complex models often fail to outperform simple baselines. Vesco et al. [40] analyze ViEWS competition results, concluding that “simplistic models with a few powerful variables performed as well as complex models.” This motivates our systematic ablation study comparing 4-feature simple models with 16-feature complex models augmented with network architecture.

2.5 Network Features in Conflict Analysis

Social network analysis in conflict research examines how actor relationships shape patterns of violence. Theoretical motivation comes from alliance politics, rebel fragmentation, and coalition dynamics literatures. We identify two streams: work that shows network features improve prediction, and work that finds null or negative effects.

2.5.1 Positive Network Findings

Dorff, Gallop, and Minhas [41, 42] demonstrate that network features—reciprocity, transitivity, and latent factor models—significantly improve the prediction of “who fights whom and when” in Nigeria. Network competition measures predict civilian victimization even controlling for conflict intensity [42]. Their work establishes that relational patterns capture dynamics standard features miss, particularly regarding targeting decisions.

Broader international relations research supports the notion of network effects. Xu, Cao, and Radil [43] find that alliance network connections suppress conflict up to three degrees of separation. Jackson and Nei [44] show that trade networks interact with military alliances to reduce the probability of war. Gade et al. [45] demonstrate that ideological homophily drives rebel collaboration networks. These studies suggest that network structure mediates conflict outcomes.

Brandt et al. [38] apply graph neural networks (ST-GCN with temporal components) to conflict forecasting, achieving improvements over ViEWS benchmarks. However, they acknowledge that “choice of metric matters significantly” — AUROC gains may not translate into operational utility.

2.5.2 Null and Negative Network Findings

Recent critical work challenges the utility of network features for forecasting. Chadeaux [46] identifies fundamental limits in conflict prediction: marginal contribution of additional variables “decreases quickly and reaches zero” regardless of sophistication. His work with Schincariol [16] provides systematic evidence that “structural covariates frequently add little or degrade performance” compared to autoregressive models.

Mueller and Rauh [35] find that standard structural variables “never add forecasting power,” and that text-based forecasts outperform models that include conventional predictors. Guimerà et al. [47] test structural balance theory in social networks, finding it fails to predict conflict between team members while simpler group-based models succeed.

Critically, the 2023/24 ViEWS Prediction Challenge [11] revealed that sophisticated ML models were often beaten by basic “no-change” baselines, particularly for onset prediction. Croicu [48] argues conflict prediction is “over-reliant on spatio-temporal features and mechanistic models,” excluding actor agency that network features theoretically capture.

2.5.3 Our Contribution: Testing Network Features for Disappearances

This mixed evidence motivates our systematic evaluation. Previous positive findings involve dynamic relational patterns (reciprocity, who-fights-whom) rather than static network structure (density, centralization). Our co-occurrence networks may fail to capture meaningful relationships if actors appearing in the same event are adversaries rather than collaborators. Additionally, disappearances may be driven by state capacity, a macro-level attribute poorly captured by monthly network metrics rather than actor-level network dynamics.

We test whether 12 graph-theoretic features (density, centralization, clustering, component structure, edge weights) improve the prediction of disappearance. Our comprehensive ablation study across nine feature configurations provides empirical evidence on the utility of network features for a previously unstudied outcome, contributing to ongoing debate about complex versus simple models in conflict forecasting.

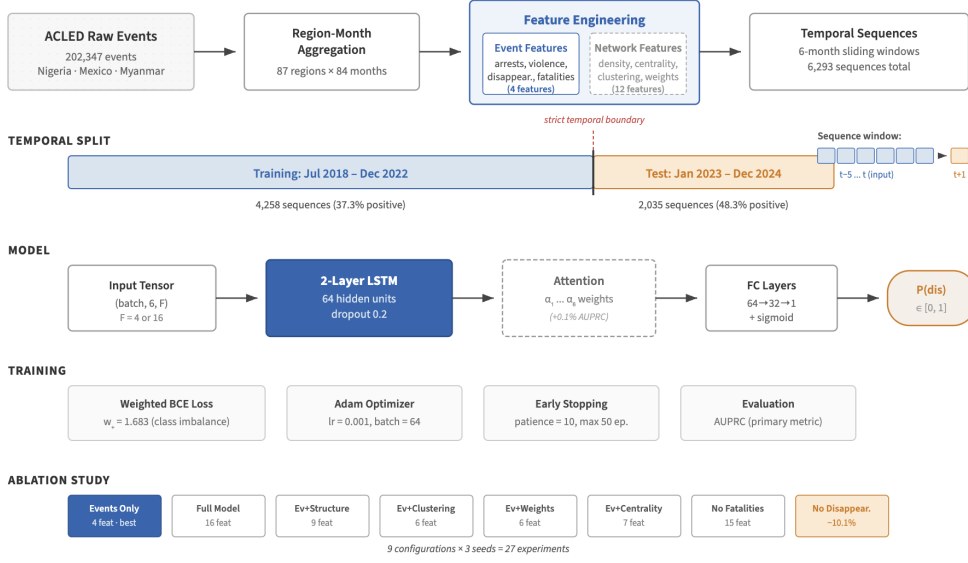


Fig. 1 End-to-end prediction pipeline. ACLED conflict events from Nigeria, Mexico, and Myanmar (2018–2024) are aggregated to region-month units and transformed into event-based features (solid border) and actor network features (dashed border; 12 metrics providing no predictive value). Six-month sliding windows produce temporal sequences for LSTM processing. The strict temporal split prevents information leakage, with training on 2018–2022 and testing on 2023–2024. Nine ablation configurations systematically test feature contributions across 27 experiments. The recommended model (Events Only, highlighted) uses 4 features to achieve AUPRC 0.745.

3 Data and Methods

This section describes our ML analysis pipeline from raw conflict events to monthly disappearance forecasts. Figure 1 provides an end-to-end overview of the workflow, which we detail in the subsections that follow.

3.1 Data Collection and Preparation

We analyze conflict event data from the Armed Conflict Location & Event Data Project (ACLED), which provides geo-located, time-stamped records of political violence and protest events worldwide [49]. ACLED codes events from diverse sources, including local and international news, humanitarian reports, and NGO documentation, with each record containing location coordinates, administrative boundaries, event type, involved actors, fatality estimates, and descriptive text. Our analysis spans January 2018 through December 2024 and covers three countries selected through a principled multi-criteria approach.

Country selection balanced four considerations. First, we required a sufficiently high prevalence of the outcome (disappearance) to enable statistical modeling. Mexico has documented a large number of disappeared persons since 1962, with 90% occurring

after 2006 during militarized anti-drug operations [21]. Myanmar experienced systematic disappearances following the February 2021 military coup, targeting political opposition and civil society [22]. Nigeria exhibits disappearances across multiple conflict zones involving both insurgent groups and state security forces. Second, we sought geographic and contextual diversity to test model generalizability across conflict types: state repression against political opposition in post-coup Myanmar, criminal violence with state collusion in Mexico’s cartel war, and complex multi-actor insurgency in Nigeria’s Northeast. Third, these countries provide substantial subnational variation, with Nigeria’s 37 administrative regions, Mexico’s 32 states, and Myanmar’s 18 administrative units ranging from peaceful areas to active conflict zones. Fourth, ACLED maintains reliable, consistent coverage for all three countries throughout our study period, verified through systematic checks for anomalous reporting gaps.

This three-country design serves as a proof-of-concept for a scalable, data-driven framework. By selecting contexts with significant variance in state capacity, perpetrator types, and conflict intensity, we evaluate the robustness of our findings across diverse empirical landscapes. This diversity allows us to move beyond case-specific idiosyncrasies to determine if the limited utility of network features is a consistent property of disappearance dynamics. Our approach enables high-granularity ablation studies that maintain computational feasibility while ensuring the results remain generalizable across different conflict archetypes. Ultimately, this multi-contextual validation demonstrates that our data-driven indicators are sensitive to the underlying mechanisms of repression rather than being artifacts of a specific geographic or political environment.

Our final dataset comprises 202,347 conflict events across 87 unique first-level administrative units (admin1 regions). These events include 5,959 enforced disappearances (2.9%), 6,913 arrests (3.4%), 114,564 violent events including battles, explosions, and violence against civilians (56.6%), and 197,817 total fatalities. Figure 2 shows the temporal distribution of disappearances across the three countries.

We use admin1-level aggregation to balance data sufficiency with policy-relevant geographic precision, following the spatial granularity employed by leading conflict forecasting systems [9].

3.2 Outcome Variable and Temporal Aggregation

Following established practices in conflict prediction [9, 14], we construct a binary outcome variable at the region-month level rather than modeling event counts. Our outcome $y_{i,t} \in \{0, 1\}$ indicates whether region i experienced at least one enforced disappearance during month t . This binary formulation prioritizes outbreak detection—identifying which regions require monitoring attention—over precise victim enumeration, aligning with operational early warning requirements [12].

This design choice reflects both substantive and methodological considerations. Operationally, early warning systems allocate limited monitoring resources across geographic units, requiring binary risk assessments rather than exact counts of cases. Substantively, disappearance counts suffer from severe underreporting, making presence/absence more reliably observed than precise magnitudes [25]. Methodologically,

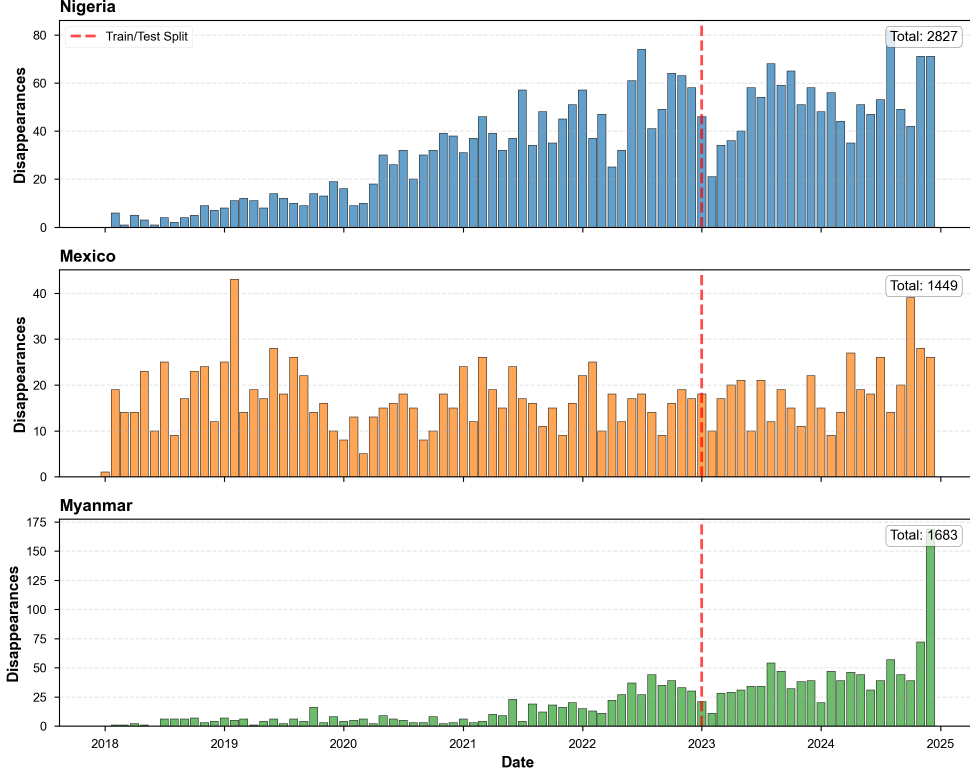


Fig. 2 Monthly disappearance events by country, 2018–2024. The vertical dashed line marks the train-test split boundary (January 2023). The test period exhibits increased disappearance prevalence (48.3% positive region-months vs. 37.3% in training), reflecting escalation across study countries.

count models face challenges with extreme zero-inflation: 97.1% of events are non-disappearances at the event level. Binary monthly aggregation concentrates this sparse signal, yielding 39.1% positive months (2,665 of 6,815 region-months) while preserving the fundamental classification challenge of distinguishing affected from unaffected regions. Furthermore, treating this problem as a regression task would likely yield misleading performance metrics; in highly sparse contexts, standard loss functions like Root Mean Square Error (RMSE) tend to favor models that predict near-zero values, effectively masking the rare outbreaks we seek to identify. By framing this as a classification task, we can use threshold-independent metrics such as AUPRC to better capture the model’s ability to distinguish signal from noise.

Table 1 Complete feature set with theoretical motivations. Event features capture directly observable conflict patterns; network features represent actor relationship structures within monthly region-specific co-occurrence networks.

Category	Feature	Theoretical Motivation
Event-Based	arrests	State repression capacity; precursor to disappearances
	violence	Overall conflict intensity; operational environment
	disappearances	Temporal persistence; campaign continuity
	fatalities	Severity of violence; escalation indicator
Structure	num_actors	Actor diversity; complexity of conflict landscape
	num_edges	Interaction extent; level of engagement
	density	Connectivity saturation; integrated vs. fragmented
	num_components	Network fragmentation; coalition structure
	largest_component	Dominant coalition size; coordination capacity
Centrality	avg_degree	Typical connectivity level; diffuse vs. concentrated
	max_degree	Most connected actor; potential dominance
	centralization	Concentration around hub; state capacity proxy
Clustering	avg_clustering	Local cohesion; neighborhood transitivity
	transitivity	Global cohesion; organized violence potential
Edge Weights	avg_edge_weight	Typical interaction intensity; sustained engagement
	max_edge_weight	Strongest pairing; repeated confrontations

3.3 Feature Engineering

We construct two feature categories capturing complementary aspects of conflict dynamics: event-based features that measure observable violence patterns, and network structure features that represent actor configurations. Table 1 summarizes all 16 features with their theoretical motivations.

3.3.1 Event-Based Features

Event features quantify conflict dynamics directly observable in ACLED records. The *arrests* feature counts events coded under ACLED’s “Strategic developments” type with “Arrests” sub-event tag, capturing state repression activity that may precede disappearances. The *violence* feature aggregates battles, explosions, and violence against civilians to measure overall conflict intensity. The *disappearances* feature counts prior disappearance events within the same region-month, representing temporal autocorrelation and campaign continuity. The *fatalities* feature sums reported deaths across all event types, indicating the severity of violence. These features reflect established findings that recent conflict history dominates prediction performance [9] and that process

variables capturing recent events can substitute for structural factors over short forecasting horizons [35]. These measures provide observable manifestations of territorial state coercive activity, enabling us to empirically evaluate institutional explanations of disappearances.

3.3.2 Network Structure Features

For each region-month, we construct an actor co-occurrence network and extract twelve graph-theoretic features. This approach builds on work demonstrating that relational patterns improve conflict prediction [41], and tests whether these benefits extend to disappearance forecasting. Networks are constructed by treating unique armed actors (ACLED codes up to 2 actors per event) as nodes, with undirected edges connecting actor pairs that appear in the same event. Edge weights reflect co-occurrence frequency. Networks are constructed independently for each region-month using only events from that spatiotemporal unit. We implement network construction and feature extraction using NetworkX [50], with edge cases (empty networks or single actors) handled by returning zero-valued feature vectors.

From these monthly networks, we compute features across four categories. *Basic structure features* quantify network size and connectivity: number of actors, number of edges, density (ratio of realized to possible connections), number of disconnected components, and largest component size. *Centrality features* measure concentration: average degree (mean connections per actor), maximum degree (most connected actor), and degree centralization [51]. *Clustering features* capture local cohesion: average clustering coefficient (neighborhood transitivity) and global transitivity (ratio of closed triads to connected triples). *Edge weight features* measure interaction intensity: average and maximum co-occurrence frequencies.

The theoretical motivation for network features rests on several mechanisms. High centralization may indicate a dominant state actor with capacity for systematic repression. Multiple disconnected components may signal fragmentation creating unstable environments conducive to violations. High clustering may reflect organized violence by cohesive actor coalitions. High edge weights indicate sustained interactions between specific actor pairs. However, as Chadeaux and Schincariol [16] demonstrate, complex features frequently fail to improve predictions over simple autoregressive models. Our ablation studies test whether these network features add predictive value specifically for disappearances.

3.4 Temporal Sequence Construction and Train-Test Splitting

We transform region-month features into fixed-length temporal sequences suitable for recurrent neural networks [36]. Each sequence comprises six consecutive months of features (months $t-5$ through t) paired with the binary outcome at month $t+1$. This sliding window approach generates one training example per region-month with sufficient history (≥ 7 months of data). For instance, predicting February 2023 disappearances in Lagos, Nigeria, uses feature vectors from August 2022 through January 2023 as input. We selected the 6-month lookback window to balance temporal

context—capturing medium-term escalation patterns beyond single-month noise—with sample size constraints, as longer windows reduce usable sequences by requiring more historical data. This design follows best practices in conflict forecasting with temporal models [17, 18].

We employ strict temporal splitting to prevent data leakage and enable genuine out-of-sample evaluation [33]. The training period spans July 2018 through December 2022 (54 months), while the test period covers January 2023 through December 2024 (24 months), approximating a 70–30 temporal split. Unlike random cross-validation, this chronological division respects causality: models trained on past data predict future outcomes. All test sequences occur strictly after all training sequences, ensuring no future information contaminates model training. This split creates a realistic forecasting scenario in which the test period exhibits a distributional shift—48.3% positive rate versus 37.3% in training—reflecting a substantive escalation in 2023–2024 disappearances across our study countries. Models must generalize across this temporal shift and demonstrate robustness to changing conflict dynamics. The temporal split and sequence construction are illustrated in the pipeline overview (Figure 1, rows 1–2). Our final dataset comprises 6,293 sequences: 4,258 training sequences (37.3% positive) and 2,035 test sequences (48.3% positive).

Before model training, we normalize features using z-score standardization to zero mean and unit variance: $z = (x - \mu)/\sigma$. Critically, normalization parameters μ and σ are computed exclusively on training data and applied to both training and test sets, preventing information leakage from test distribution into preprocessing [52]. Post-normalization, the training set exhibits a mean of 0.000 and a standard deviation of 1.000 by construction, while test statistics differ, reflecting the distributional shift.

3.5 Model Architecture

We evaluate two Long Short-Term Memory (LSTM) architectures: a standard LSTM baseline and an attention-augmented variant. Both models employ identical recurrent processing but differ in how they aggregate temporal information for final classification.

The standard LSTM processes input sequences of shape (batch size, 6 timesteps, F features) through a 2-layer stacked LSTM with 64 hidden units per layer and dropout rate 0.2. At each timestep t , the LSTM computes hidden state $h_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1})$ where x_t is the feature vector, h_{t-1} is the previous hidden state, and c_{t-1} is the cell state. After processing all six timesteps, the final hidden state h_6 passes through a classification head: a fully connected layer reducing dimensionality from 64 to 32 with ReLU activation and dropout 0.2, followed by an output layer projecting to a single logit with sigmoid activation producing probability $\hat{y} \in [0, 1]$. This architecture contains approximately 58,000 parameters.

The attention-LSTM extends this architecture by learning which months contribute most to predictions [39]. Rather than using only h_6 , it computes attention weights over all hidden states $h_1, \dots, h_6$. For each timestep t , we compute attention score $e_t = W_2 \cdot \tanh(W_1 \cdot h_t)$ where W_1 and W_2 are learned weight matrices. These scores are normalized via softmax: $\alpha_t = \exp(e_t) / \sum_{i=1}^6 \exp(e_i)$, producing attention weights that sum to one. The context vector $c = \sum_{t=1}^6 \alpha_t h_t$ aggregates all hidden

states weighted by their importance. This context vector then passes through the same classification head as the standard LSTM. The attention mechanism adds approximately 2,000 parameters, yielding $\approx 60,000$ total. Beyond potential performance gains, attention provides interpretability: if $\alpha_6 \gg \alpha_1$, predictions rely primarily on the most recent month, while uniform weights indicate equal contribution across the lookback window. Both architectures are depicted in the model row of Figure 1.

3.6 Training Configuration and Optimization

We train models using binary cross-entropy loss with class weighting to address the 37.3% positive, 62.7% negative imbalance in training data [31]:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [w_+ \cdot y_i \log(\hat{y}_i) + w_- \cdot (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where $w_+ = (1 - p)/p = 0.627/0.373 = 1.683$ and $w_- = 1.0$. This weighting penalizes false negatives 1.683 times more severely than false positives, compensating for the underrepresentation of disappearances in training data.

We optimize using Adam [53] with learning rate 0.001, batch size 64, and default momentum parameters ($\beta_1=0.9$, $\beta_2=0.999$). Training runs for a maximum of 50 epochs with early stopping: if test set AUPRC does not improve for 10 consecutive epochs, training terminates. Regularization comes exclusively from dropout [54] applied at rate 0.2 in both LSTM layers and fully connected layers; we do not employ L2 weight decay. All models train on CPU in 2–3 minutes per random seed using PyTorch 2.0, as the modest model size ($\approx 60,000$ parameters) and dataset size ($\approx 6,000$ sequences) make GPU acceleration unnecessary.

3.7 Evaluation Metrics

Following best practices for imbalanced classification [32], we report multiple complementary metrics with primary emphasis on the Area Under the Precision-Recall Curve (AUPRC). The precision-recall curve plots precision ($TP/(TP + FP)$) versus recall ($TP/(TP + FN)$) across all decision thresholds, focusing exclusively on positive class performance. AUPRC summarizes this curve as a single scalar, with higher values indicating better classification.

We prioritize AUPRC over AUROC for three reasons. First, precision and recall depend only on true positives, false positives, and false negatives, making them insensitive to the large number of true negatives in our 48.3% positive test set. AUROC’s false-positive rate includes true negatives in its denominator, potentially yielding optimistic estimates when the majority class dominates. Second, random classifier performance differs meaningfully across metrics: AUPRC equals the positive class prevalence (0.483 in our test set), providing an informative baseline, whereas AUROC always equals 0.50 regardless of class balance. Third, for operational early warning systems, precision-recall tradeoffs directly correspond to policy-relevant questions about acceptable false alarm rates versus detection rates.

As secondary metrics, we report AUROC for comparison with prior work, Brier score measuring probability calibration ($\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$, lower is better), and precision-recall values at specific operating points for deployment scenario analysis.

3.8 Ablation Study Design

To identify which features drive predictive performance, we conduct comprehensive ablation studies testing nine feature configurations, with each configuration trained using three random seeds (42, 123, 456) for a total of 27 experiments. Table 2 specifies all configurations and their purposes.

Table 2 Ablation study configurations. All experiments employ identical training procedures (same seeds, hyperparameters, temporal split, early stopping) for fair comparison. Feature counts vary by configuration while total sequences (6,293) remain constant.

Configuration	# Feat.	Purpose
Full Model	16	Baseline with all features
Events Only	4	Quantify network contribution
Network Only	12	Quantify event contribution
No Disappearances	15	Measure autocorrelation dependency
No Fatalities	15	Test fatalities marginal value
Events + Structure	9	Isolate basic structure metrics
Events + Centrality	7	Isolate centrality metrics
Events + Clustering	6	Isolate clustering metrics
Events + Weights	6	Isolate edge weight metrics

We maintain strict experimental control across all ablations. Each configuration uses identical random seeds, ensuring the same weight initializations and batch orderings. Hyperparameters remain fixed: learning rate 0.001, batch size 64, dropout 0.2, class weights 1.683/1.0. The train-test temporal split is identical (2018–2022 train, 2023–2024 test). Early stopping uses the same criterion: 10 epochs without test AUPRC improvement. This control isolates feature composition as the sole experimental variable.

Our analysis strategy compares specific configuration pairs to answer targeted questions. Comparing Full Model versus Events Only quantifies network feature contribution by measuring performance change when removing all twelve network features. Comparing Full Model versus No Disappearances measures autocorrelation dependency, revealing how much performance derives from temporal persistence versus other predictive signals. Comparing Events Only versus each Events + [category] configuration tests whether specific network feature categories (structure, centrality, clustering, weights) add value when combined with event features. We report results as mean $\pm$ standard deviation across three random seeds. With only three replicates, we emphasize effect sizes and consistency across runs rather than formal significance testing. Differences exceeding one standard deviation and consistent across all three seeds are interpreted as meaningful.

We additionally compare LSTM models to logistic regression to assess whether temporal modeling provides value over linear feature combinations. The logistic regression baseline uses identical features flattened to a single vector ($6 \text{ timesteps} \times F \text{ features dimensions}$), the same train-test temporal split, and class weighting (1.683 for positives), but employs default scikit-learn settings without hyperparameter tuning. If LSTM substantially outperforms logistic regression, it indicates that temporal structure—captured by recurrent connections across the 6-month lookback window—provides genuine predictive signal beyond linear combinations of lagged features [28].

3.9 Reproducibility

To facilitate replication and extension, we will release complete code, environment specifications, and trained models upon publication. Our GitHub repository will contain Python implementations of all preprocessing pipelines, feature engineering, model architectures, training procedures, and evaluation scripts, with exact package versions specified (PyTorch 2.0.1, NumPy 1.24.3, Pandas 2.0.2, NetworkX 3.1, scikit-learn 1.3.0). All experiments use fixed random seeds (42, 123, 456) for deterministic replication. ACLED data are publicly available at <https://acleddata.com/> with free registration;

This research uses publicly available conflict event data and does not involve human subjects. All analysis focuses on aggregate regional patterns rather than individual identification. Code and data sharing follows ACLED’s terms of use permitting academic research with proper attribution.

4 Results

We present results in three stages: model architecture comparison, systematic feature ablations, and country-level performance. All metrics represent mean $\pm$ standard deviation across three random seeds (42, 123, 456). Figure 3 provides visual overview of ablation results.

4.1 Model Architecture Comparison

Table 3 compares LSTM architectures against baselines on the full 16-feature set.

Table 3 Performance comparison of model architectures on the full 16-feature set. LSTM models trained with 3 random seeds; logistic regression with a single run. Random baseline equals test set prevalence (48.3%). Gap = Test – Train AUPRC.

Model	Test AUPRC	Train AUPRC	Gap	Brier
Attention-LSTM	0.741 ± 0.003	0.705 ± 0.003	-0.036	0.208 ± 0.001
Standard LSTM	0.740 ± 0.003	0.701 ± 0.004	-0.039	0.206 ± 0.001
Logistic Regression	0.717	0.626	-0.091	0.212
Random Baseline	0.483	—	—	—

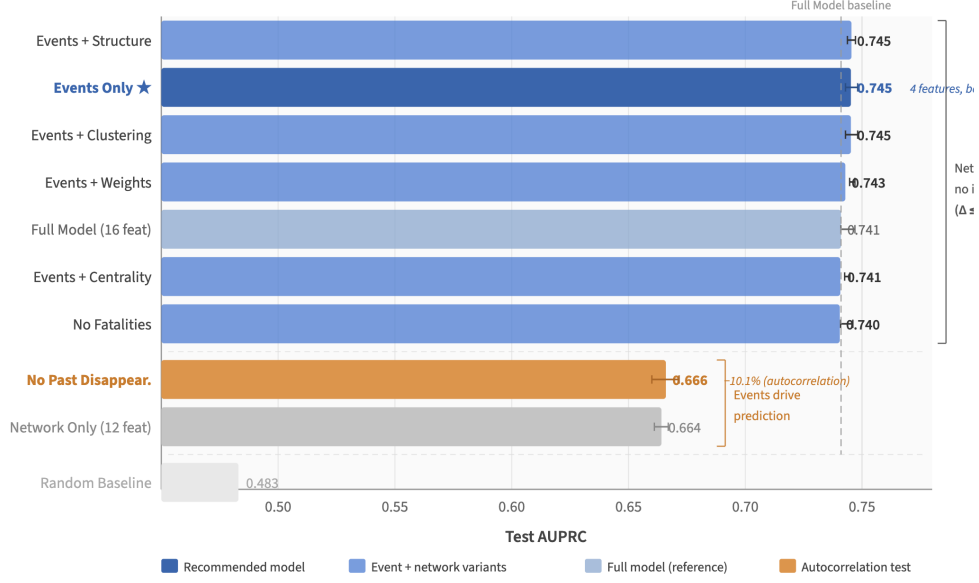


Fig. 3 Feature ablation study results sorted by test AUPRC (mean $\pm$ SD across 3 random seeds). The Events Only model (★, 4 features) matches or outperforms all network-augmented configurations, including the Full Model with 16 features. Adding any network feature category provides no meaningful improvement (all $\Delta \leq \pm 0.004$). The dashed vertical line marks the Full Model baseline. Removing past disappearances causes the largest single-feature drop (-10.1%), but the model retains substantial predictive power (0.666) well above random (0.483), demonstrating early warning capability for new outbreak regions.

Both LSTM models achieve test AUPRC ≈ 0.74 , representing 25.8-point improvement over random baseline and 2.4-point improvement over logistic regression. This confirms temporal modeling captures predictive structure beyond linear feature combinations. Attention-LSTM marginally outperforms Standard LSTM (0.741 vs 0.740), a difference within measurement noise. We adopt Attention-LSTM for subsequent ablations to enable temporal interpretability. The negative train-test gaps reflect distributional shift: test set contains 48.3% positive region-months versus 37.3% in training.

4.2 Feature Ablation Results

Table 4 presents systematic ablations testing nine feature configurations.

Results show three distinct performance tiers visible in Table 4 and Figure 3. The top tier (ranks 1–7, AUPRC 0.740–0.745) comprises all configurations including the four event-based features, regardless of network features added or removed. The middle tier (ranks 8–9, AUPRC 0.664–0.666) contains configurations missing past disappearances or lacking event features. The bottom tier is random baseline (0.483). This structure indicates event-based features, particularly past disappearances, drive predictive performance.

Table 4 Feature ablation study results sorted by test AUPRC. Δ indicates change relative to the Full Model (16-feature) baseline. #F = number of input features. All experiments use Attention-LSTM with identical seeds, hyperparameters, and temporal split.

Rank	Configuration	Test AUPRC	Δ	#F
1	Events + Structure	0.745 ± 0.002	+0.004	9
2	Events Only	0.745 ± 0.003	+0.004	4
3	Events + Clustering	0.745 ± 0.003	+0.004	6
4	Events + Weights	0.743 ± 0.001	+0.002	6
5	Full Model	0.741 ± 0.003	—	16
6	Events + Centrality	0.741 ± 0.001	−0.000	7
7	No Fatalities	0.740 ± 0.003	−0.001	15
8	No Disappearances	0.666 ± 0.006	−0.075	15
9	Network Only	0.664 ± 0.003	−0.077	12
<i>Random Baseline</i>		<i>0.483</i>	<i>−0.258</i>	—

Three key findings emerge from ablations. First, events-only outperforms the full model, indicating twelve network structure features provide no predictive value. If actor network configuration were a primary driver of disappearances, removing network features would materially reduce predictive performance. We observe no such reduction. This holds when examining individual network categories: structure features contribute +0.0 points, clustering +0.0, weights −0.002, centrality −0.005. Events-only also outperforms network-only by 8.1 points despite using one-third the features, confirming simple conflict indicators contain far more information than complex network metrics.

Second, removing past disappearances reduces AUPRC from 0.741 to 0.666 (7.5-point drop, 10.1% of performance), confirming it as the strongest single predictor. However, the model without disappearance history remains 18.3 points above random, meaning arrests, violence, and fatalities alone provide substantial predictive power for identifying new outbreak regions.

Third, removing fatalities causes negligible change (0.741 to 0.740), suggesting fatality counts are redundant with violence event counts.

4.3 Recommended Model

Based on ablation results, we recommend the Events Only Attention-LSTM as optimal configuration:

This model achieves marginally better performance (+0.004 AUPRC) with 75% fewer features, eliminates computationally expensive network construction, provides interpretable predictions for stakeholders, and is more robust to data quality variation.

4.4 Country-Level Performance

Table 6 shows performance is relatively consistent across three distinct conflict contexts (range 0.736–0.758).

Table 5 Recommended model specification achieving best test performance with fewest features.

Property	Specification
Architecture	2-layer Attention-LSTM, 64 hidden units
Features	arrests, violence, disappearances, fatalities
Feature count	4 (vs 16 for Full Model)
Test AUPRC	0.745 ± 0.003
Parameters	$\approx 55,000$

Table 6 Country-level performance of the Events Only model.

Country	Regions	Test seq.	Pos. rate	AUPRC
Nigeria	37	740	41.2%	0.758
Mexico	32	672	53.6%	0.741
Myanmar	18	623	50.4%	0.736
Overall	87	2,035	48.3%	0.745

The 2.2-percentage-point spread across fundamentally different conflict environments (state repression in Myanmar, criminal violence with state collusion in Mexico, multi-actor insurgency in Nigeria) suggests temporal features capture generalizable precursors rather than context-specific patterns.

5 Discussion

This study demonstrates that enforced disappearances can be predicted despite extreme rarity (AUPRC 0.745, more than double random baseline), that simple event-based features outperform complex network-augmented models, and that models retain predictive power (0.666 AUPRC) without past disappearances, indicating genuine early warning capability. We discuss substantive interpretation, practical and theoretical implications, and limitations.

5.1 Why Network Features Fail

Network features consistently fail despite theoretical motivation. We propose four complementary explanations. First, disappearances are predominantly state-directed repression where institutional capacity matters more than actor network structure. In Mexico, state-cartel collusion drives disappearances [21]; in Myanmar, security forces systematically target opposition [22]; in Nigeria, both insurgents and state forces commit violations. Network metrics capturing fragmentation or coalition structure may be relevant for inter-group violence [41], but disappearances require institutional capacity poorly captured by monthly actor co-occurrence patterns.

Second, our co-occurrence networks are crude proxies: actors appearing in the same event may be adversaries not collaborators, actor names contain inconsistencies, and many disappearances involve unnamed state agents absent from records. Third, network features may be redundant with event counts: regions with more violence mechanically produce networks with more actors and edges. Fourth, disappearances may be driven by slow-moving structural conditions (regime type, judicial weakness) operating on longer timescales than monthly network snapshots, which reflect measurement noise rather than meaningful dynamics.

The consistency across all network categories, seeds, and the combined Full Model suggests co-occurrence network structure genuinely fails to capture disappearance mechanisms, aligning with evidence that complex features frequently degrade conflict forecasting [16, 40].

5.2 What Drives Disappearance Prediction

Past disappearances contribute 10.1% of performance, confirming violations occur in sustained campaigns rather than isolated incidents, consistent with qualitative research on escalation within frameworks of impunity [19]. Conflict intensity—captured by violence counts and arrest events—provides the remaining 89.9%, supporting arguments that process variables can substitute for structural factors over short horizons [15]. Violence and arrests create the operational environment for disappearances: security operations produce detainees, confrontations provide cover for extrajudicial actions, and instability weakens accountability. The near-zero contribution of fatalities specifies this mechanism: frequency of state-civilian contact (event counts) predicts disappearances more than lethality.

Actor network structure is notably absent from prediction. Who is present matters far less than what happened, suggesting disappearances are driven by the intersection of state capacity and conflict intensity rather than network dynamics.

5.3 Implications for Early Warning and Methodology

The events-only model’s performance (0.745 AUPRC) using four features enables operational deployment: monthly counts of arrests, violence, disappearances, and fatalities are reliably coded in ACLED, require no network construction, and are interpretable to stakeholders. The model predicts new outbreaks without prior history of disappearance (AUPRC = 0.666), addressing operational needs beyond extrapolation from existing patterns. However, a one-month prediction horizon limits preventive action, admin1 aggregation lacks precision for targeted protection, and models provide risk scores rather than actionable intelligence about perpetrators or tactics.

Methodologically, simple features outperforming complex networks reinforce patterns across recent conflict forecasting: sophisticated models frequently underperform simple baselines [11], autoregressive models outperform complex alternatives [16], and simplistic models with few powerful variables perform as well as complex ones [40]. This suggests the marginal value of complexity decreases rapidly once core signals are

captured. Our autocorrelation ablation provides a template: quantify persistence contribution (10.1%) while demonstrating residual predictive power (89.9%), balancing acknowledgment of temporal dependence with genuine forecasting capability.

5.4 Theoretical and Substantive Implications

Our feature importance analysis indicates that enforced disappearances align more closely with institutionally driven phenomena than with actor-network dynamics. The minimal predictive utility of network structure features—despite theoretical motivation from violence diffusion literature [1, 2]—suggests a boundary condition for network-centric explanations. Instead, the data offers empirical support for state capacity theories [3–5], where disappearances are driven by macro-level structural conditions and institutional support systems—such as command chains and concealment infrastructure—that appear poorly captured by monthly actor co-occurrence patterns. These results suggest a scope condition for network-centric theories: while relational structure may explain patterns of inter-group violence, disappearances appear primarily shaped by territorially embedded state coercive processes.

The results of temporal modeling (LSTM outperforming logistic regression) suggests non-linear temporal dynamics, with the model capturing medium-term escalation patterns. While much predictive signal remains accessible to linear combinations consistent with known conflict processes [16], this empirical trend carries three primary implications:

First, the dominance of event-based features (arrests, violence counts) indicates that disappearances follow territorial state presence and conflict intensity. This represents the intersection of where states possess the capacity to conceal violations and where conflict provides operational cover, rather than being a product of relational configurations among armed actors.

Second, the 10.1% contribution of past disappearances points toward a degree of bureaucratic persistence, where repressive capacity tends to persist through institutional inertia once established. However, since the majority of predictive performance is derived from conflict intensity indicators, disappearances do not appear to be purely self-perpetuating. They remain highly responsive to the operational context—such as arrest surges—characterizing them as institutionally-mediated responses to conflict dynamics.

Third, the model’s ability to predict new outbreaks at AUPRC 0.666 highlights a significant predictive signal in contextual risk factors. This suggests that the measurable indicators of institutional capacity for concealment can manifest rapidly when conflict intensity crosses specific thresholds. This finding complements structural accounts by suggesting that repressive capacity can be activated quickly given sufficient operational pressure, even without a prior history of violations.

From a data science perspective, the cross-contextual robustness observed across Nigeria, Mexico, and Myanmar suggests that these patterns represent stable empirical regularities. This consistency indicates that our model captures general mechanisms of disappearance production, providing a scalable foundation for monitoring specific violation types across diverse high-intensity conflict settings.

5.5 Limitations

ACLED captures only reported disappearances; underreporting is likely severe in remote regions and repressive contexts where reporting itself is dangerous. This selection bias means models learn to predict reported disappearances, which may differ from actual violations. However, this distinction is less problematic for early warning applications where monitoring organizations operate within the same information environment.

Our three-country design trades analytical depth for geographic breadth: comprehensive ablation studies (nine configurations, multiple seeds) rather than superficial multi-country analysis. This enables rigorous testing that network features provide no value, distinguishing genuine null effects from noise. However, external validity is limited. All three countries exhibit high-intensity disappearance contexts where mechanisms may differ from low-prevalence settings; state actors or state-affiliated groups dominate, potentially under-representing purely non-state disappearances; and shared characteristics (weak judicial institutions, limited accountability, ongoing conflict) may not generalize. Critical questions require validation: Do network features also fail in civil war contexts (Syria, Yemen) where rebel coalitions may matter more? Do findings hold in post-conflict transitions (Colombia)? Does the dominance of simple features extend to lower-prevalence contexts (Turkey, Philippines)?

Methodological choices constrain findings. The six-month lookback may miss longer-term patterns; one-month horizon limits prevention; admin1 aggregation obscures within-region variation; binary outcomes discard count information. Our co-occurrence networks are crude proxies: co-occurrence may indicate adversaries not collaborators, actor names contain inconsistencies, and many disappearances involve unnamed agents. Alternative constructions (adversarial ties, temporal decay weighting, bipartite graphs) might yield different results. Our negative finding applies to static structural metrics from co-occurrence networks, not all possible network representations. The 2018–2024 temporal scope provides only 24-month out-of-sample testing; longer-term stability and response to structural changes (peace agreements, regime transitions) remain unknown.

5.6 Future Work

Priority validation should test findings across broader country samples: civil war contexts where actor coalitions may matter more (Syria, Yemen), post-conflict transitions targeting specific populations (Colombia), state repression without insurgency (Belarus, Iran), and lower-intensity settings with sporadic disappearances (Philippines, Turkey). Multi-country studies could employ hierarchical modeling to pool information while estimating country-specific effects, meta-learning to identify generalizable versus context-dependent findings, or transfer learning testing whether models trained on high-data countries predict in low-data contexts.

Additional extensions include longer prediction horizons (3–6 months) for earlier intervention, subnational granularity (admin2, grid-cell) for geographic precision, text-based features from event descriptions capturing contextual nuances, dynamic network

features tracking temporal evolution of targeting patterns, and incorporation of slow-moving structural variables (regime type, judicial independence) as cross-sectional complements. Deployment pilots with human rights organizations would provide the most meaningful validation of operational utility.

6 Conclusions

Enforced disappearances are institutionally driven phenomena reflecting state capacity to conceal violence rather than actor network dynamics. This finding—derived from ML as a theory-testing tool—adjudicates between competing explanations by showing that network structure features (density, centralization, clustering) do not meaningfully improve prediction beyond simple indicators of territorial control and conflict intensity. This negative result is theoretically informative: disappearances occur where states can conceal violations and where conflict provides operational cover (arrests, violence), not primarily through relational configurations among armed actors.

Our analysis identifies three social mechanisms underlying disappearances. First, bureaucratic persistence: past disappearances contribute 10.1% of predictive performance, indicating that violations often unfold as sustained campaigns reflecting institutional inertia within security forces. Second, contextual responsiveness: 89.9% of predictive power derives from conflict intensity (arrests, violence), suggesting disappearances remain highly sensitive to the operational environment despite institutional persistence. Third, rapid capacity activation: models predict new outbreaks (AUPRC 0.666 without disappearance history), implying that repressive infrastructure can emerge quickly once violence crosses critical thresholds, challenging accounts that assume long-term institutional development is required.

These patterns hold across Nigeria, Mexico, and Myanmar—contexts that differ in regime type and conflict structure yet exhibit a common dynamic in which disappearances follow state presence and conflict intensity rather than network configurations. This cross-national consistency points to general mechanisms in high-intensity settings, though validation in civil war contexts with stronger non-state actors remains an important next step.

Methodologically, we demonstrate that severe human rights violations (2.9% of 202,347 events) are computationally tractable prediction targets, achieving AUPRC 0.745. The recommended model’s simplicity—four event-based features: arrests, violence, past disappearances, and fatalities—supports operational feasibility by relying only on routinely coded conflict data without network construction. This parsimony reinforces growing evidence that additional feature complexity often yields limited returns once core predictive signals are captured.

For policy, interventions should prioritize institutional accountability and territorial governance rather than actor network disruption. Early warning systems can detect outbreak risk from escalating arrests and violence before disappearance patterns become entrenched, supporting preventive action. However, short prediction horizons and data limitations mean computational risk scores must complement, not replace, contextual intelligence. More broadly, this study illustrates how ML can evaluate social theories through systematic feature comparison. For enforced disappearances,

what happened—and where state capacity operates—matters far more than who was involved.

Declarations

Data Availability

The ACLED conflict event data used in this study are publicly available at <https://acleddata.com/> with free registration for academic use. All code for data preprocessing, feature engineering, model training, and evaluation is available at <https://github.com/christoschr97/ed-temporal-network> under an MIT License. Trained model checkpoints are available from the corresponding author upon reasonable request for research purposes.

Competing Interests

The authors declare that they have no competing interests.

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Author Contributions

All authors contributed equally to this work. CJC, CD and NI: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft and Review & Editing.

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